The usage of AI

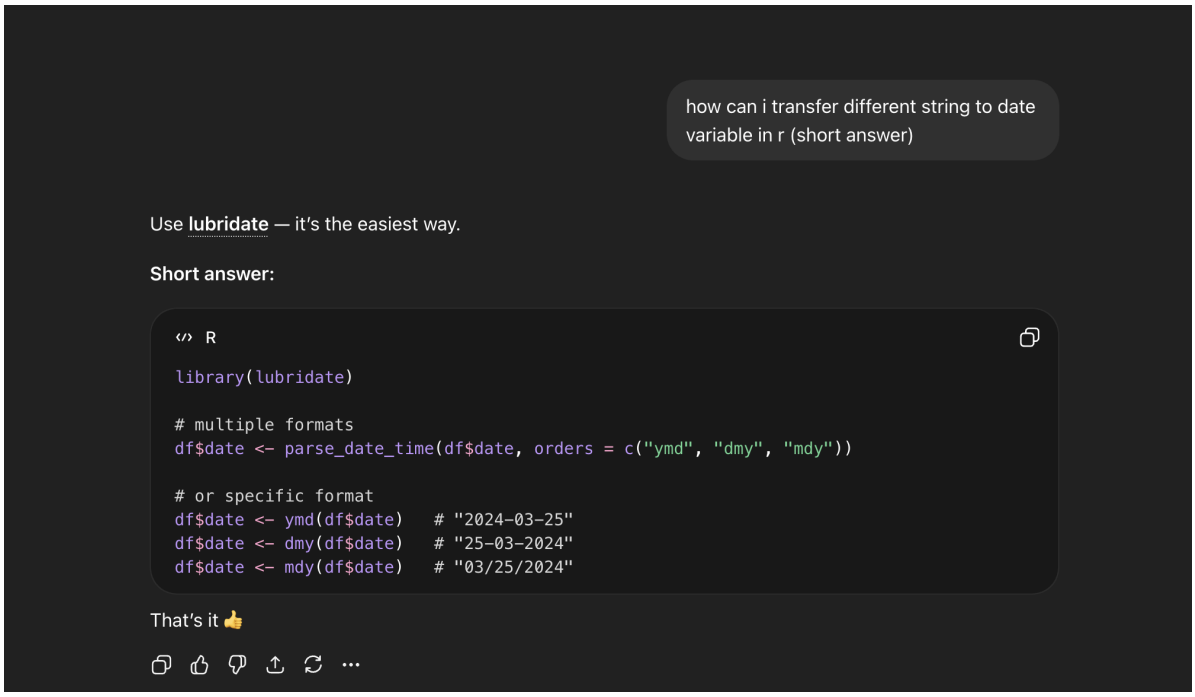


Figure 1: R code for transfering to date

how can i color different group when i use ggplot

Use the `color` or `fill` aesthetic in **ggplot2**.

Short answer:

1. For points / lines → use `color`

```r
ggplot(data, aes(x = x_var, y = y_var, color = group_var)) +
  geom_point()
```

2. For bars / boxplots → use `fill`

```r
ggplot(data, aes(x = group_var, y = value, fill = group_var)) +
  geom_boxplot()
```

3. Custom colors (optional)

```r
+ scale_fill_manual(values = c("red", "blue", "green"))
```

👉 Key idea:

- `color` = outline / points / lines
- `fill` = inside of shapes (bars, boxes)

If you want, I can fix your exact plot so it looks nicer 👍

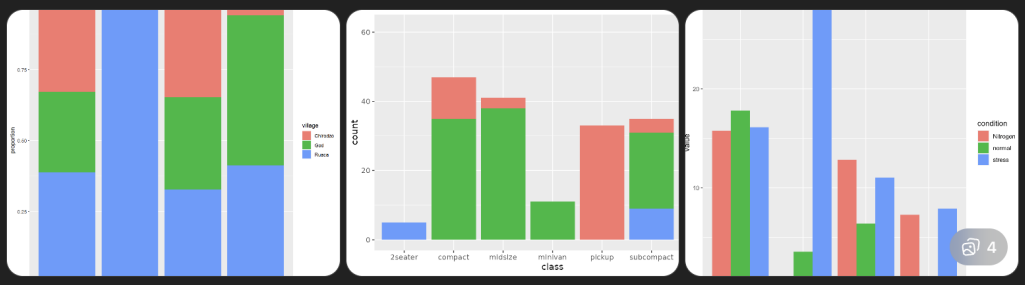
Figure 2: R code for group colors in plot

what are the types of plot in ggpolt

Here are the **main types of plots** in ggplot2 (short + useful for your assignment):

📊 Common plot types

1. Bar plot (counts / categories)




```R
ggplot(data, aes(x = category)) +
  geom_bar()
```

👉 Use for: number of observations per group (e.g. segment count)

Figure 3: Types of ggplot

Figure 4: Types of ggplot

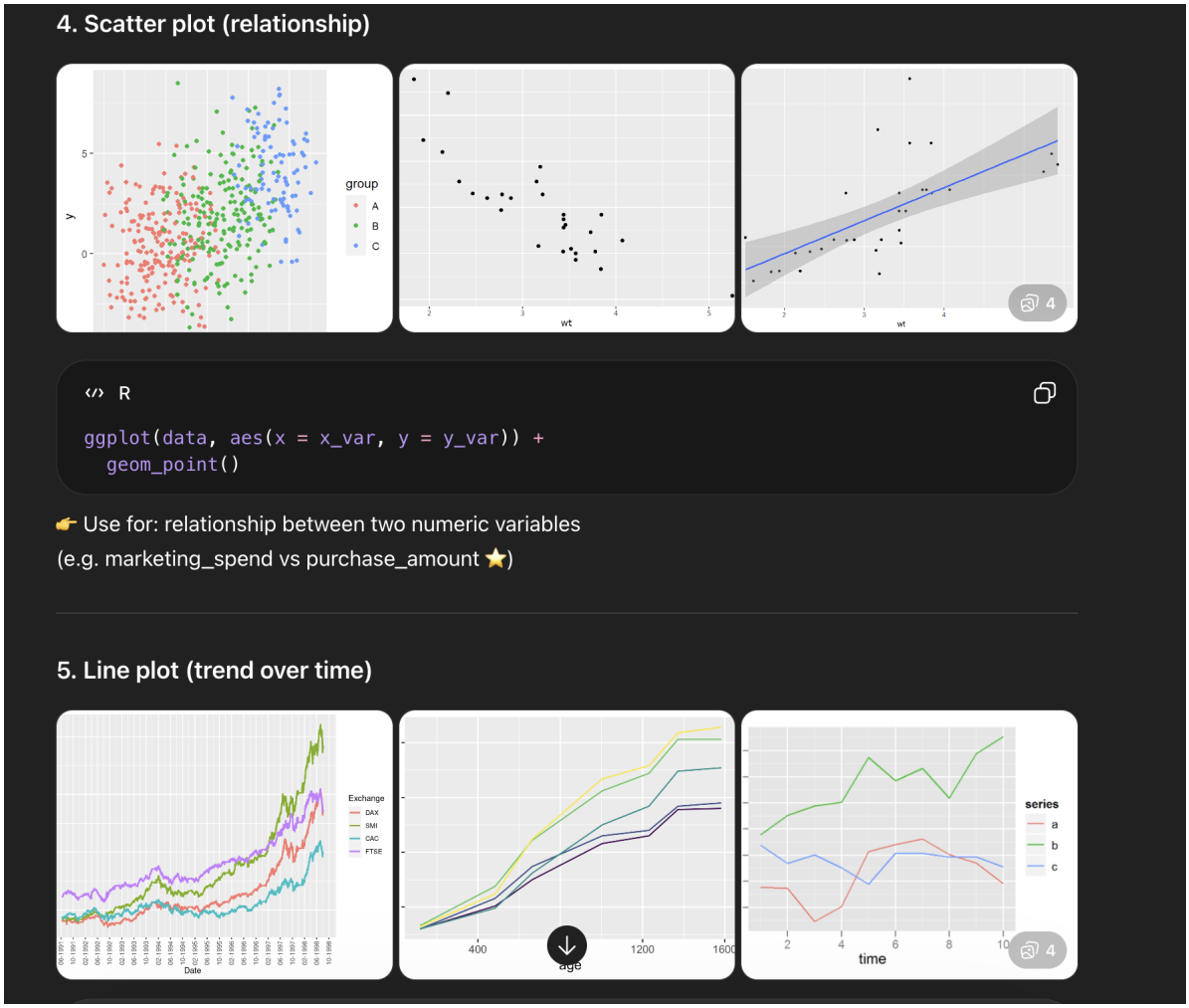


Figure 5: Types of ggplot

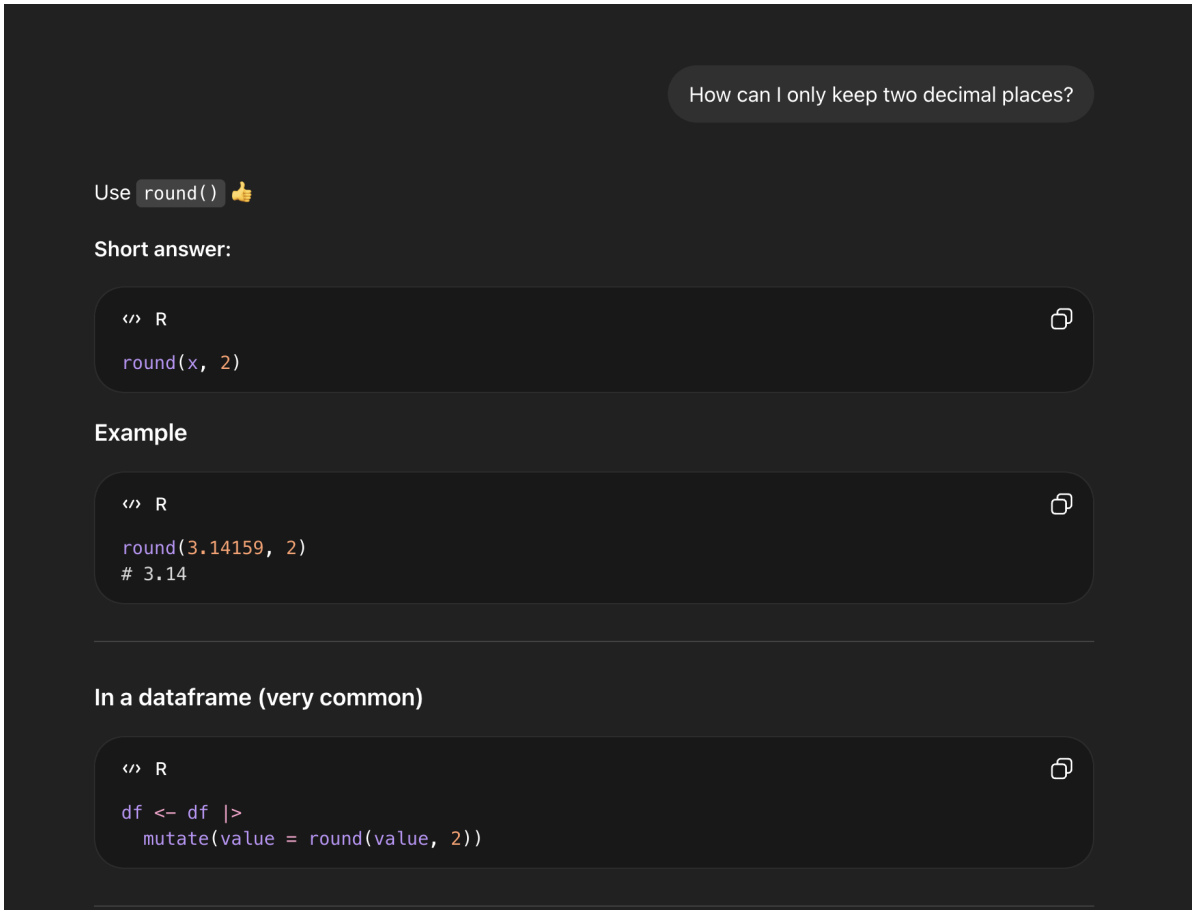


Figure 6: R code for save decimal place

how can i change the scale of y-axis(zoom in only)

Use `coord_cartesian()` 👍 (this is the correct way to *zoom in only* in **ggplot2**)

✅ Zoom in (recommended)

```r
ggplot(data, aes(x, y)) +
  geom_point() +
  coord_cartesian(ylim = c(0, 100))
```

👉 This:

- **zooms the view only**
- **keeps all data** (important for boxplots, stats, etc.)

❌ Don't use this for zoom

```r
ylim(0, 100)
```

👉 This **removes data outside the range** (can mess up your plot)

Figure 7: R code for zoom y-axis

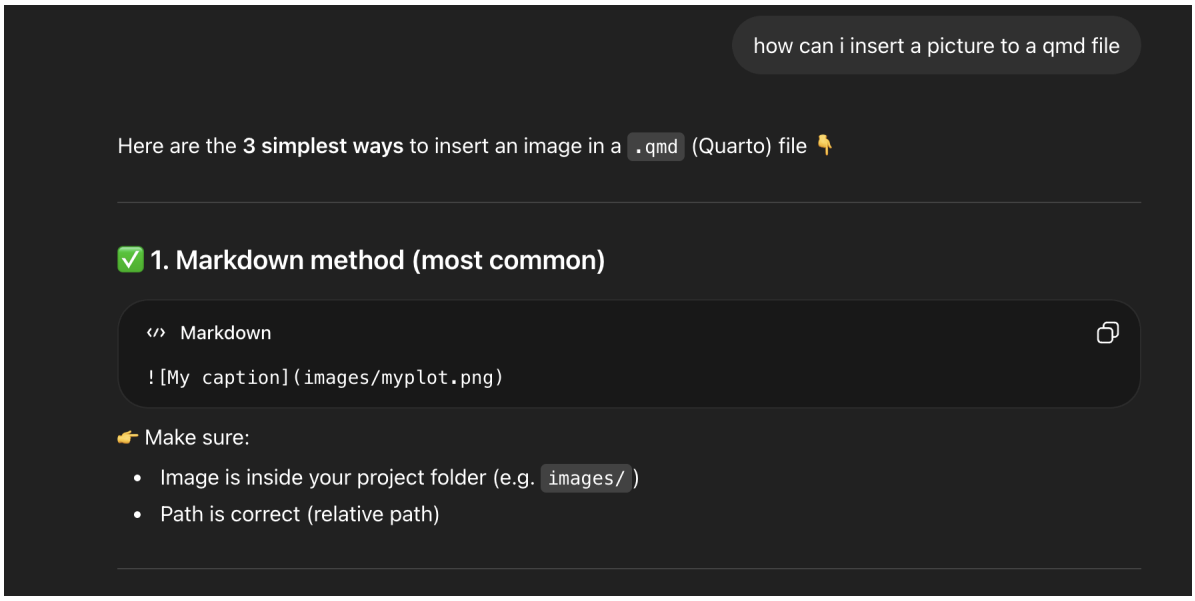


Figure 8: R code for insert picture

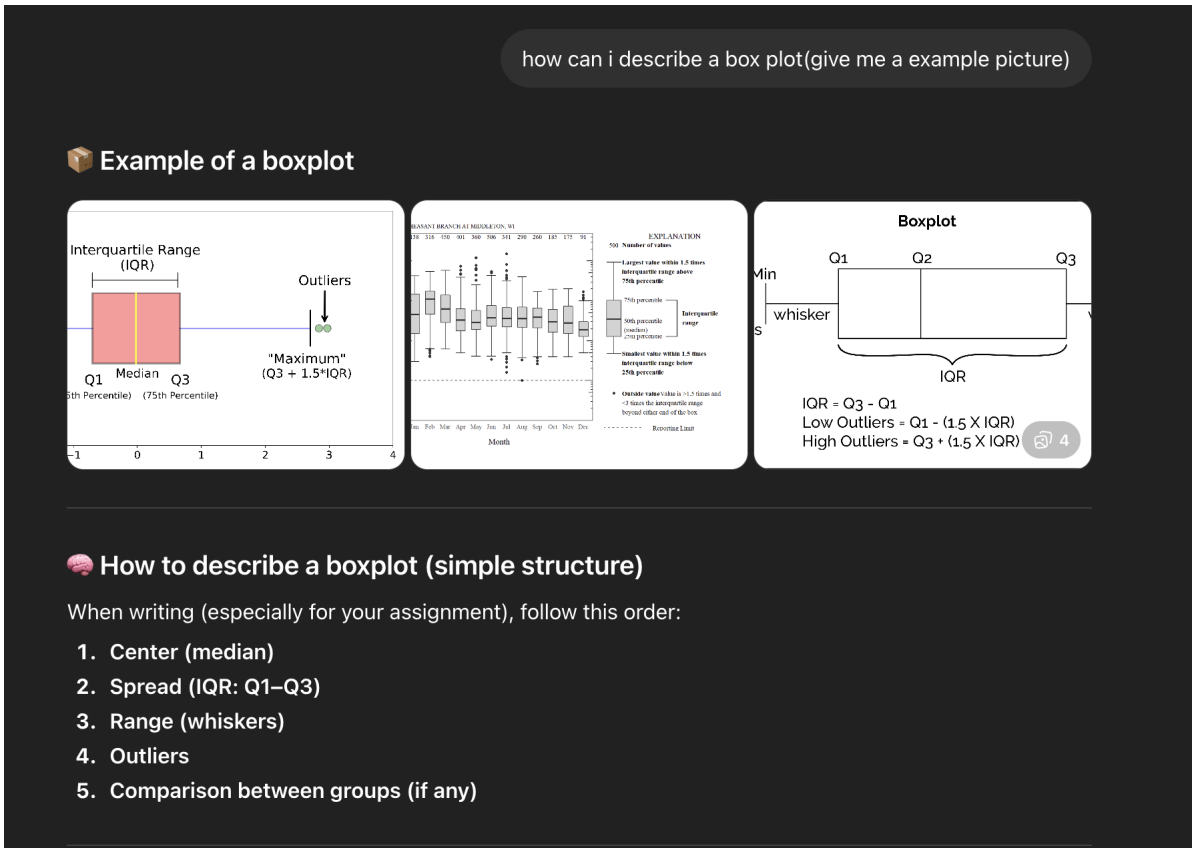


Figure 9: help to describe boxplot

how to plot a linear regression

Here's the **quick way** to plot a linear regression in **ggplot2** 👇

✅ 1. Basic linear regression plot

R

```r
ggplot(data, aes(x = x_var, y = y_var)) +
  geom_point() +
  geom_smooth(method = "lm")
```

👉 This gives:

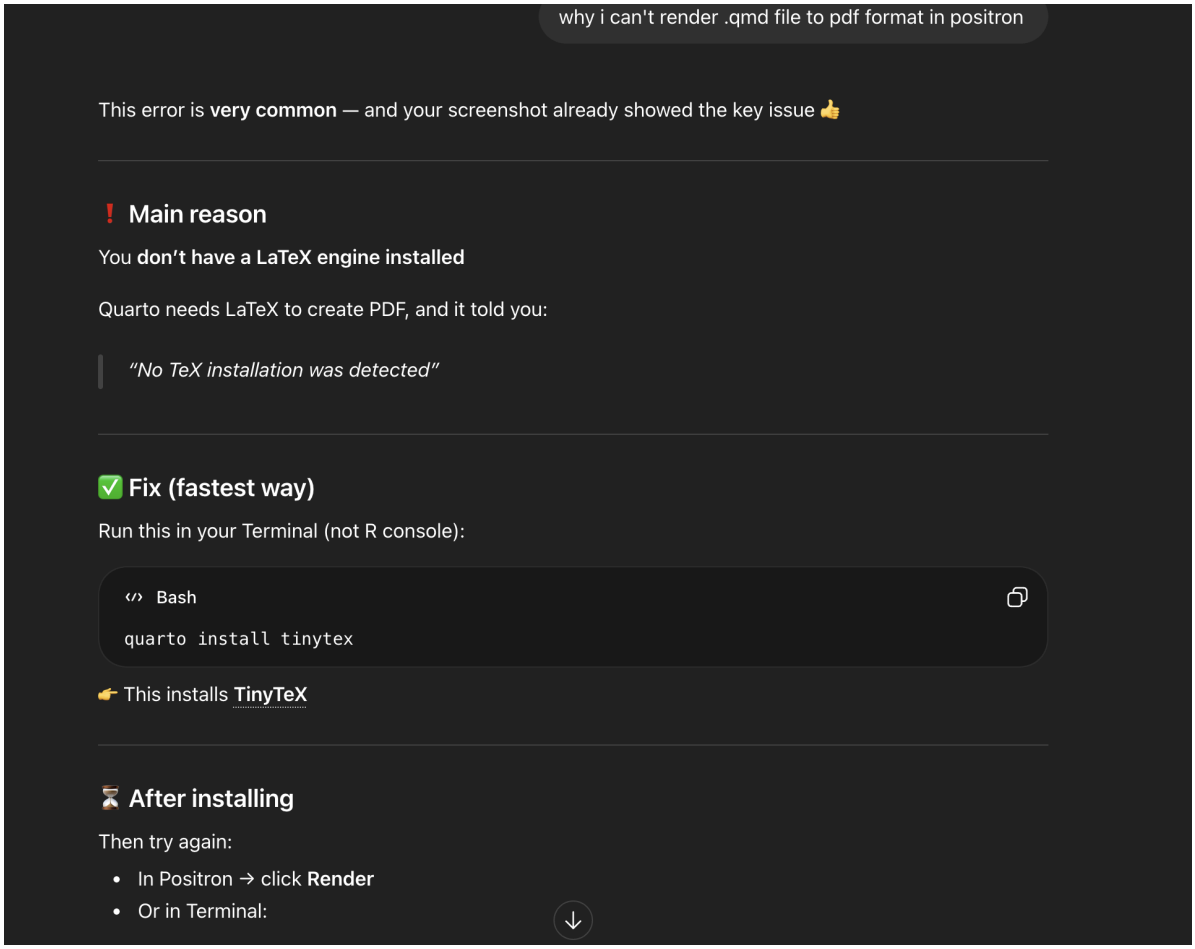
- scatter plot
- • regression line (automatically)

✅ 2. Remove shaded confidence interval (cleaner)

R

```r
ggplot(data, aes(x = x_var, y = y_var)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

Figure 10: R code for plot linear regression



why i can't render .qmd file to pdf format in positron

This error is **very common** — and your screenshot already showed the key issue 👍

❗ Main reason

You **don't have a LaTeX engine installed**

Quarto needs LaTeX to create PDF, and it told you:

> *"No TeX installation was detected"*

✅ Fix (fastest way)

Run this in your Terminal (not R console):

```bash
quarto install tinytex
```

👉 This installs **TinyTeX**

⏳ After installing

Then try again:

- In Positron → click **Render**
- Or in Terminal:

Figure 11: help to render to pdf

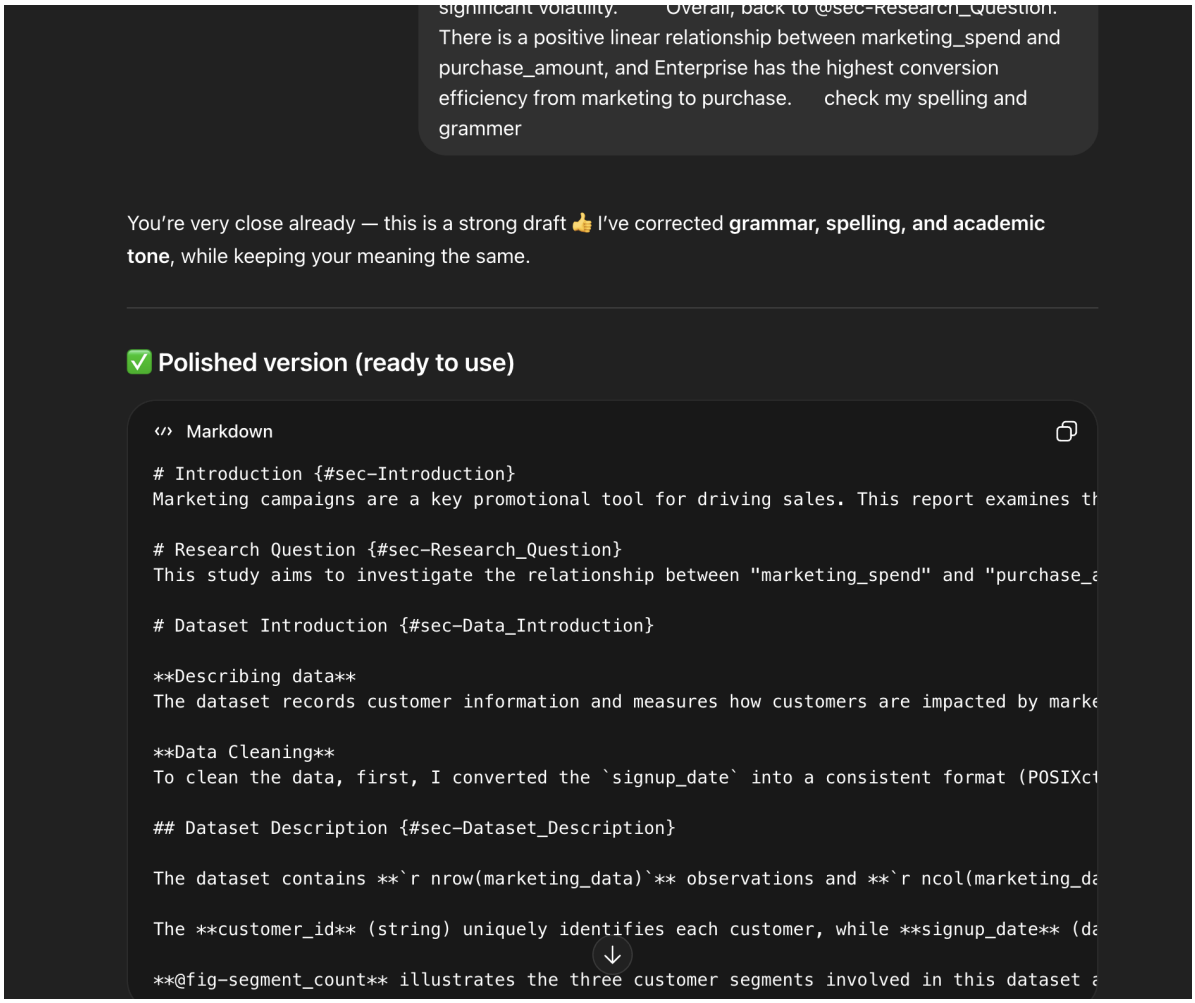


Figure 12: help to check the English